Photosynthesis and Diffusion

1. **Question:** The cells inside a plant leaf use up carbon dioxide when they photosynthesise. Use what you have learnt about diffusion to explain how carbon dioxide diffuses into the leaf.
   Answer: Carbon dioxide diffuses into the leaf through tiny pores called stomata. Because the plant constantly uses up carbon dioxide for photosynthesis, there is less carbon dioxide inside the leaf than in the air outside. The gas moves down this concentration gradient, naturally spreading from the higher concentration outside to the lower concentration inside.

2. **Question:** The cells inside a plant leaf produce oxygen when they photosynthesise. Suggest what happens to this oxygen.
   Answer: The plant uses some of this oxygen itself to carry out respiration, which gives it energy. The remaining, extra oxygen diffuses out of the leaf through the stomata and goes back into the air.

Minerals and Plant Growth

3. **Question:** All organisms need protein for growth. Compare the way in which plants obtain protein with the way that you and other animals obtain protein.
   Answer: Plants make their own proteins by combining the glucose they produce during photosynthesis with nitrates they absorb from the soil through their roots. Animals cannot make proteins from scratch like this; they must get their protein by eating plants or other animals.

4. **Question:** It can be difficult, even for an expert, to tell the difference between a plant that is short of magnesium, and one that is short of protein. Suggest why.
   Answer: Both magnesium and nitrogen (which plants use to make protein) are essential for making chlorophyll, the green pigment in leaves. If a plant lacks either of these minerals, it cannot make enough chlorophyll, causing the leaves to turn yellow. Because the visible symptom is the same, it is hard to tell which mineral is missing just by looking.

The Carbon Cycle (Food Chains)

1. (a) **Question:** Draw a food chain with one plant and three animals in it.
   Answer: Grass → Grasshopper → Frog → Snake

   (b) **Question:** The arrows in your food chain represent energy passing from one organism to the next. Do they also show how carbon atoms pass from one organism to the next? Explain your answer.
   Answer: Yes, the arrows also show the flow of carbon atoms. Living things are made of carbon-containing compounds like carbohydrates, proteins, and fats. When one animal eats another organism, it consumes these compounds, meaning the carbon atoms are passed along the food chain.

2. **Question:** The human body contains atoms of many different elements. Carbon is one of the most common elements. Name three different compounds in your body

that contain carbon atoms.
Answer: Carbohydrates (such as glucose), proteins, and fats (lipids).

The Carbon Cycle (Models and Predictions)

1. **Question:** In this model, all the carbon atoms in one place move to another place at the same time. Is that what happens in the real carbon cycle? If not, how could you modify this model to make it a better representation? If you can, try out your suggestion on your model.
   Answer: No, in the real carbon cycle, atoms do not all move at once; it is a continuous and gradual process happening all the time. To make the model better, you could move small numbers of atoms one by one to show a steady, continuous flow rather than moving them all in one big group.

2. **Question:** Predict what would happen to the carbon atoms if one of the processes stopped completely - for example, combustion. If you can, try it out on your model.
   Answer: If combustion stopped, the rapid transfer of carbon from fossil fuels into the atmosphere as carbon dioxide would stop. The carbon would remain safely stored underground as coal, oil, and gas, which would slow down the increase of carbon dioxide in the air.

3. **Question:** If you drew a carbon cycle to show what was happening before humans were present on Earth, how would it differ from the carbon cycle diagram above?
   Answer: A diagram from before humans existed would not include the combustion of fossil fuels by factories, cars, or power plants. The cycle would consist entirely of natural processes like photosynthesis, respiration, and decomposition.

4. **Question:** Explain why fossil fuels are non-renewable resources.
   Answer: Fossil fuels are non-renewable because they take millions of years to form from the buried remains of ancient plants and animals under intense heat and pressure. Humans are burning them much faster than nature can ever replace them.